INCH-POUND
MIL-DTL-24640/9E
8 November 2011
SUPERSEDING
MIL-DTL-24640/9D
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, -20 TO +105 °C, 600 VOLTS, LIGHTWEIGHT, MULTI-PAIR, 22 AWG, TYPE 2XS AND 2XSW (INCLUDING VARIATION TYPES 2XSO AND 2XSOW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24640.

Construction, Non-Watertight (2XS or 2XSO) and Watertight (2XSW or 2XSOW

First	-	19-strand, tin-coated copper conductor, 22 AWG. Watertight constructions shall be constructed with 7 strands.
Second	-	Component wires shall be SAE-AS81044/12, except identification shall be in accordance with the codes and methods of MIL-DTL-24640. Testing shall be in accordance with the component wire ratings and requirements listed on this specification sheet. Conformance testing shall be in accordance with SAE-AS81044 prior to cabling.
Third	-	Two conductors, one black and one white, twisted together. The pitch shall be 1½±¼ inch.
Fourth	-	Optional binder tape.
Fifth	-	Braided shield of 38 AWG tin-coated copper strands. Braid coverage shall be no less than 90 percent determined in accordance with WC-27500.
Sixth	-	Shield insulation of two polyester tapes or a polyester tape and extruded jacket. Identification codes applied using Method 2.
Seventh	-	The required number of shielded, twisted pairs shall be cabled together. Cabling sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used, as necessary, to form a round cable.
Eighth	-	Optional binder tape.

Construction Variant for Type 2XS/W

Ninth	-	Cross-linked polyolefin jacket (see table I for thickness). NOTE: Manufacturer's identification tape or thread is required (see 3.7.3).

Construction Variant for Type 2XSO/W

Ninth	-	Braided shield, tin-coated copper strands. Braid must be constructed to meet surface transfer impedance requirements as listed herein.

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Tenth - Optional binder tape.

Eleventh - Cross-linked polyolefin jacket (see table II for thickness).
NOTE: Manufacturer's identification tape or thread is required (see 3.7.3).

TABLE I. Construction details for variant 2XS/W.

PIN M24640/9	Type number	Number of pairs	Jacket thickness (inch) (min.)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max.)
				(min.)	(max.)	
-01UN	2XS-2	2	0.038	0.308	0.332	61.4
-02UN	2XS-3	3	0.038	0.325	0.350	74.2
-03UN	2XS-7	7	0.041	0.423	0.455	135.0
-04UN	2XS-10	10	0.044	0.537	0.579	194.0
-05UN	2XS-14	14	0.044	0.582	0.627	245.0
-06UN	2XS-19	19	0.044	0.644	0.694	319.0
-07UN	2XS-24	24	0.047	0.758	0.818	431.0
-08UN	2XS-30	30	0.047	0.804	0.866	500.0
-09UN	2XSW-2	2	0.038	0.310	0.335	79.0
-10UN	2XSW-3	3	0.038	0.350	0.380	109.0
-11UN	2XSW-7	7	0.041	0.475	0.505	204.0
-12UN	2XSW-10	10	0.044	0.575	0.610	295.0
-13UN	2XSW-14	14	0.044	0.635	0.675	387.0
-14UN	2XSW-19	19	0.044	0.725	0.765	513.0
-15UN	2XSW-24	24	0.047	0.815	0.855	642.0
-16UN	2XSW-30	30	0.047	0.915	0.955	805.0

TABLE II. Construction details for variant 2XSO/W - overall shield.

PIN M24640/9	Type number	Number of pairs	Jacket thickness (inch) (min.)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max.)
				(min.)	(max.)	
-01UO	2XSO-3	3	0.038	0.380	0.410	117.0
-02UO	2XSO-7	7	0.041	0.474	0.510	190.0
-03UO	2XSO-10	10	0.044	0.594	0.640	287.0
-04UO	2XSO-14	14	0.044	0.636	0.686	323.0
-05UO	2XSO-19	19	0.044	0.709	0.765	423.0
-06UO	2XSO-30	30	0.047	0.869	0.937	644.0
-07UO	2XSOW-3	3	0.038	0.400	0.430	153.0
-08UO	2XSOW-7	7	0.041	0.500	0.540	267.0
-09UO	2XSOW-10	10	0.044	0.600	0.640	350.0
-10UO	2XSOW-14	14	0.044	0.675	0.720	460.0
-11UO	2XSOW-19	19	0.044	0.750	0.805	600.0
-12UO	2XSOW-30	30	0.047	0.925	0.975	895.0

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REQUIREMENTS:

Qualification required.

Construction details: See tables I and II.

Component wire: The component wire shall be in accordance with SAE-AS81044 and shall be tested in accordance with the following wire ratings and additional requirements.

Temperature rating	150 °C (302 °F) (max.) conductor temperature
Voltage rating	600 volts (root mean square) at sea level
Accelerated aging	
Non-blocked	Oven temperature, 300±2 °C (572±3.6 °F) for 6 hours
Water-blocked	Oven temperature, 250±2 °C (482±3.6 °F) for 6 hours
Blocking	150±2 °C (302±3.6 °F)
Flammability	30 seconds (max.)
	3.0 inches (max.)
	No flaming of tissue paper
Humidity resistance	5000 megohms for 1000 feet (min.) insulation resistance after humidity exposure
Dielectric test (100% test)	
Primary insulation	Impulse test - 6.0 kV (peak) or 3 kHz spark test - 1.5kV root mean square
Component wire	Impulse test - 8.0 kV (peak) or 3 kHz spark test - 5.5kV root mean square
Insulation resistance	5000 megohms for 1000 feet (min.)
Life cycle	
Non-blocked	Oven temperature, 200±2 °C (392±3.6 °F) for 168 hours
Water-blocked	Oven temperature, 200±2 °C (392±3.6 °F) for 120 hours
Physical properties of primary insulation	
Tensile strength	2500 lb/in^2 (min.)
Elongation	150 percent (min.)
Shrinkage	
Non-blocked	0.125 inch (max.) at 300±2 °C (572±3.6 °F)
Water-blocked	0.125 inch (max.) at 250±2 °C (482±3.6 °F)
Surface resistance	500 megohms inches (min.), initial and final readings
Thermal shock	Oven temperature, 150±2 °C (302±3.6 °F)
Maximum change in measurement	0.060 inch

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Wet dielectric test	2500 volts (root mean square)
Wrap test	Mandrel wrap test required

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	17.37
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor and shield	1000
Conductor to shield	Not applicable
Shield to shield (component shield only)	200
Component shield to overall shield (Type 2XSO/W only)	500
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor and shield	2500
Shield to shield (component shield only)	100
Shield to shield (component shield to overall shield, 2XSO/W only)	100
Conductor and shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure
Capacitance:	
Mutual capacitance (at 1 MHz, picofarads/foot (pF/ft), max.)	50
Capacitance unbalance (percent, max.)	15
Characteristic impedance (at 1 MHz, ohms)	40 (min.), 63 (max.)
Watertightness (Type 2XSW and 2XSOW only) (see table IX of MIL-DTL-24640 for limits of water leakage)	No failure
Weight	No failure

Group B:

Cold bending	No failure
Cross-link proof test (percent, max.)	
Jacket	50
Drip (Type 2XSW and 2XSOW only)	No failure
Tear (lb/in thickness, min.)	35
Physicals (unaged)	
Jacket	

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Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Attenuation (at 3 MHz, dB/100 feet, max.)	3
Gas flame (1 hour)	Not applicable

Group C:

Physicals (aged), air oven	
Jacket	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (conductor insulation, cycles, min.)	Not applicable
Permanence of printing (jacket, cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable sealant removability (Type 2XSW and 2XSOW only)	No failure
Shrinkage	No failure
Surface transfer impedance (Type 2XSO and 2XSOW only)	
Milliohms per meter (max.)	700
EMP response (dB, min.)	60

Group D:

Flame propagation	No failure

Qualification Inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	

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Jacket	25
Fillers	45
Insulation	10
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Component wire	No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
All types and sizes 2 - 14	500
All types and sizes 19 - 30	250

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2009-077)

Review Activities:
Army – AR, AV, CR, CR4
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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